

NATIONAL SCHOOL OF ARTS AND CRAFTS ENSAM Meknes

Industrial Engineering - Artificial Intelligence & Data Science

GraphPlag: A Semantic Graph-Based Plagiarism Detection System with Integrated AI-Generated Content Identification

Big Data Academic Project

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Academic Year 2025-2026

ABSTRACT

This report presents **GraphPlag**, a comprehensive plagiarism detection system that employs semantic graph-based analysis combined with AI-generated content identification. The system transforms documents into sentence-level graphs and computes similarity using both interpretable graph kernels and learned graph neural network embeddings. GraphPlag addresses the limitations of traditional text-matching approaches by detecting paraphrased and cross-lingual plagiarism through structural and semantic graph comparison. Additionally, it integrates multiple AI-detection strategies (statistical, linguistic, and neural) into a unified ensemble model. The system provides multiple interfaces (web, REST API, CLI), supports scalable deployment via Docker and Kubernetes, and generates comprehensive reports. Experimental results demonstrate that GraphPlag achieves superior performance in detecting semantically equivalent content compared to traditional methods while maintaining practical computational efficiency.

Keywords: plagiarism detection, graph-based analysis, semantic similarity, AI-generated content, graph kernels, graph neural networks, ensemble methods

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1 Introduction

1.1 Problem Statement and Motivation

The proliferation of digital content and the advancement of natural language generation models have created significant challenges in academic integrity and content originality verification. Traditional plagiarism detection systems primarily rely on text-matching algorithms that fail to identify:

- Paraphrased content where surface text differs but semantic meaning remains identical
- Cross-lingual plagiarism where content is translated between languages
- AI-generated content that exhibits different statistical and linguistic patterns than human writing

The need for more sophisticated detection mechanisms has become increasingly urgent with the widespread availability of powerful language models capable of generating human-like text.

1.2 Research Objectives

This research aims to develop a comprehensive plagiarism detection system that:

1. Detects semantic plagiarism through graph-based document representation
2. Identifies AI-generated content using multiple detection strategies
3. Provides interpretable results with actionable recommendations
4. Supports scalable deployment for practical applications
5. Offers multiple interfaces for diverse user requirements

1.3 Contributions

The primary contributions of this work include:

- A novel graph-based document representation capturing semantic relationships
- An ensemble approach combining graph kernels and graph neural networks
- Integrated AI-content detection with weighted risk assessment
- A production-ready system with multiple deployment options
- Comprehensive evaluation on multiple plagiarism detection tasks

1.4 Report Structure

This report is organized as follows: Section 2 reviews related work in plagiarism detection and AI-content identification. Section 3 details the methodology, including graph construction and similarity computation. Section 4 describes the system implementation. Section 5 presents experimental results and analysis. Section 6 discusses implications and limitations. Section 7 concludes with future research directions.

2 Related Work

2.1 Traditional Plagiarism Detection

Traditional plagiarism detection systems can be categorized into:

- **Text-matching approaches:** Employ string matching algorithms like Rabin-Karp or fingerprinting techniques
- **Stylometric analysis:** Analyze writing style features such as vocabulary richness and sentence structure
- **Citation analysis:** Examine citation patterns and reference networks
- **Vector space models:** Represent documents as term-frequency vectors and compute cosine similarity

These methods, while effective for verbatim copying, struggle with semantic plagiarism where text is substantially rephrased.

2.2 Graph-Based Methods

Graph-based approaches have shown promise in various NLP tasks:

- **Dependency parsing graphs:** Represent syntactic relationships between words
- **Knowledge graphs:** Capture semantic relationships between entities
- **Graph kernels:** Compute similarity between graph structures (Weisfeiler-Lehman, random walk)
- **Graph neural networks:** Learn graph representations through neural message passing

Previous research has demonstrated the effectiveness of graph representations for document similarity, but comprehensive systems integrating multiple graph approaches remain limited.

2.3 AI-Generated Content Detection

Current AI-detection approaches include:

- **Statistical methods:** Analyze perplexity, burstiness, and token frequency distributions
- **Linguistic features:** Examine syntactic patterns, discourse markers, and coherence metrics
- **Neural classifiers:** Train supervised models on human vs. AI-generated text pairs
- **Watermarking techniques:** Embed detectable patterns during text generation

Most existing systems address either plagiarism detection or AI-content identification, but not their integration.

3 Graph Neural Networks: Theoretical Foundations

3.1 What is a Graph?

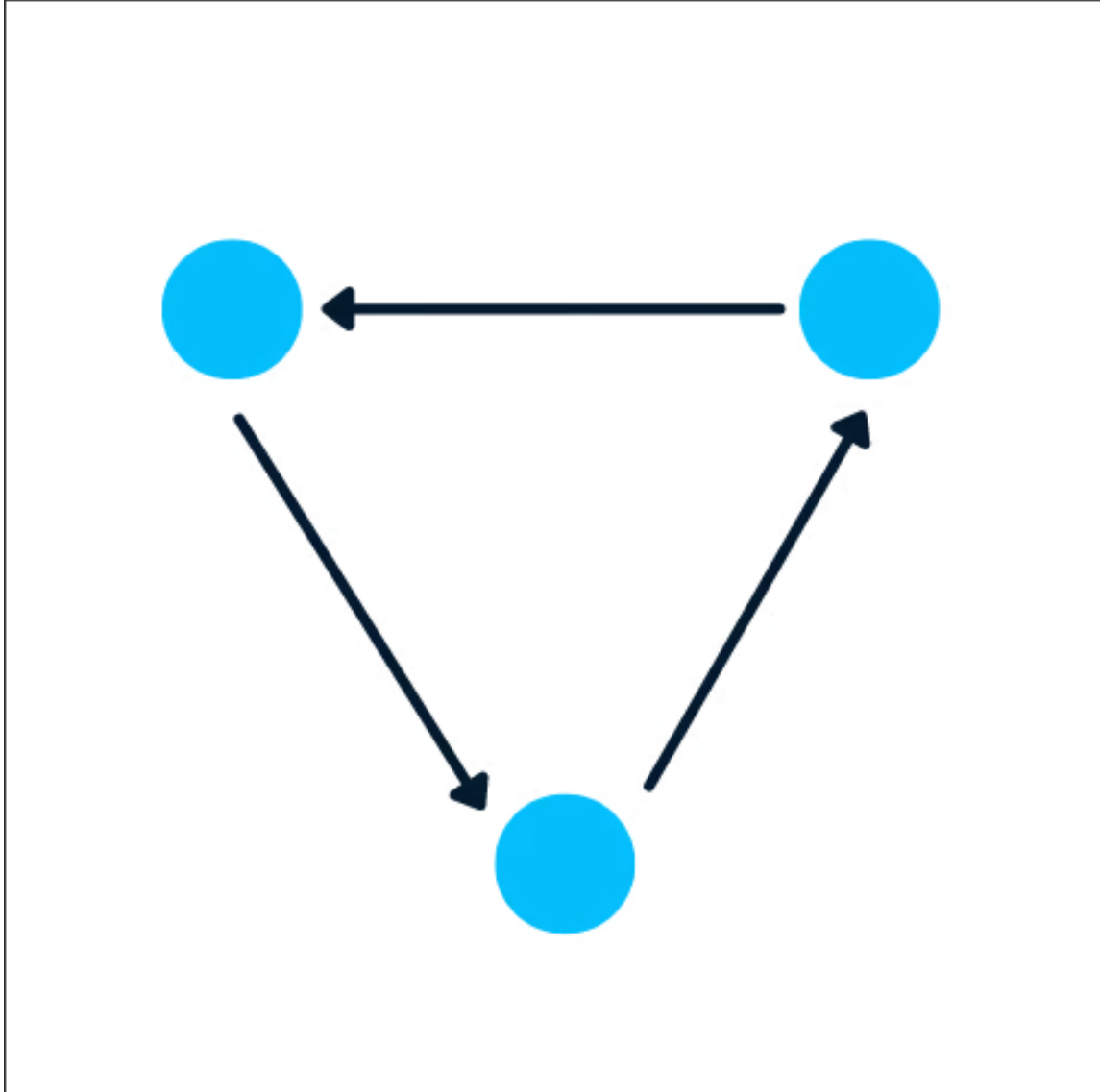


Figure 1: A simple graph

A Graph is a data structure composed of nodes (vertices) and edges. Nodes represent entities such as people, places, or objects, while edges define relationships between them. Graphs can be directed (edges have specific direction) or undirected (bidirectional relationships). Graphs excel at modeling complex relationships and interactions, making them valuable for social network analysis, recommendation systems, and semantic analysis.

3.2 What is a Graph Neural Network (GNN)?

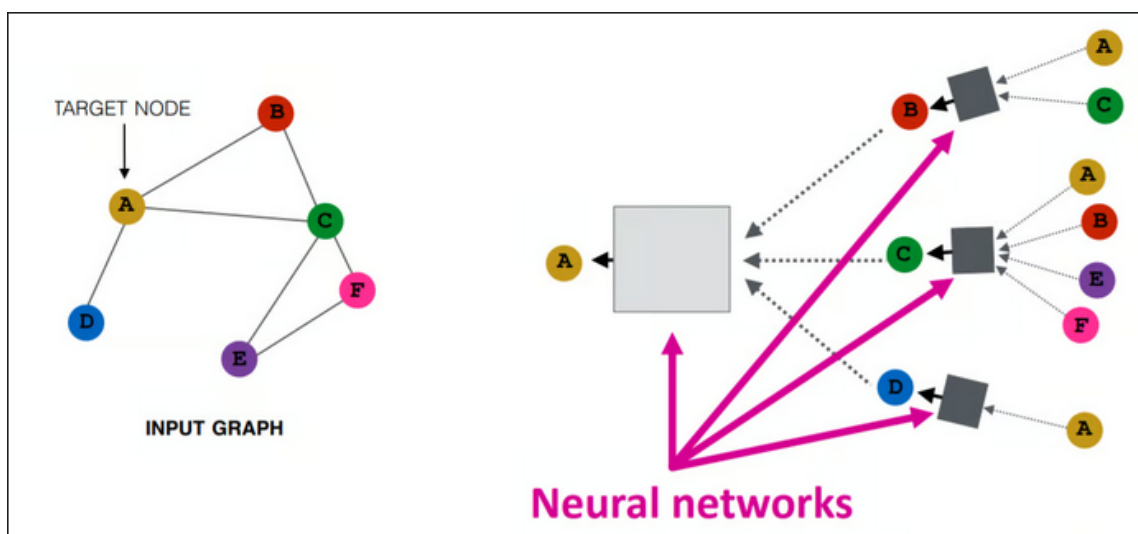


Figure 2: Graph Neural Network

Graph Neural Networks are neural networks specifically designed to process graph-structured data. Unlike traditional neural networks that work with grid-like (images) or sequential (text) data, GNNs handle the irregular, non-Euclidean structure of graphs. GNNs were developed when conventional neural networks failed on graph data due to variable graph sizes and complex structures.

The core operation in GNNs is message passing: nodes aggregate information from their neighbors, update their representations, and pass messages to connected nodes. This allows GNNs to learn representations that capture both node features and graph structure.

3.3 Key GNN Architectures

- **Graph Convolutional Networks (GCNs):** Apply convolution operations to graphs by aggregating neighboring node information. They come in two types: Spatial GCNs (direct neighborhood aggregation) and Spectral GCNs (using graph Laplacian).
- **Graph Attention Networks (GATs):** Introduce attention mechanisms to weigh neighbor contributions differently.
- **GraphSAGE:** Samples and aggregates features from node neighborhoods, enabling inductive learning on unseen graphs.

3.4 Types of Graph Neural Networks Tasks

Graph Neural Networks can be applied to various tasks involving graph-structured data. The main categories of GNN tasks include:

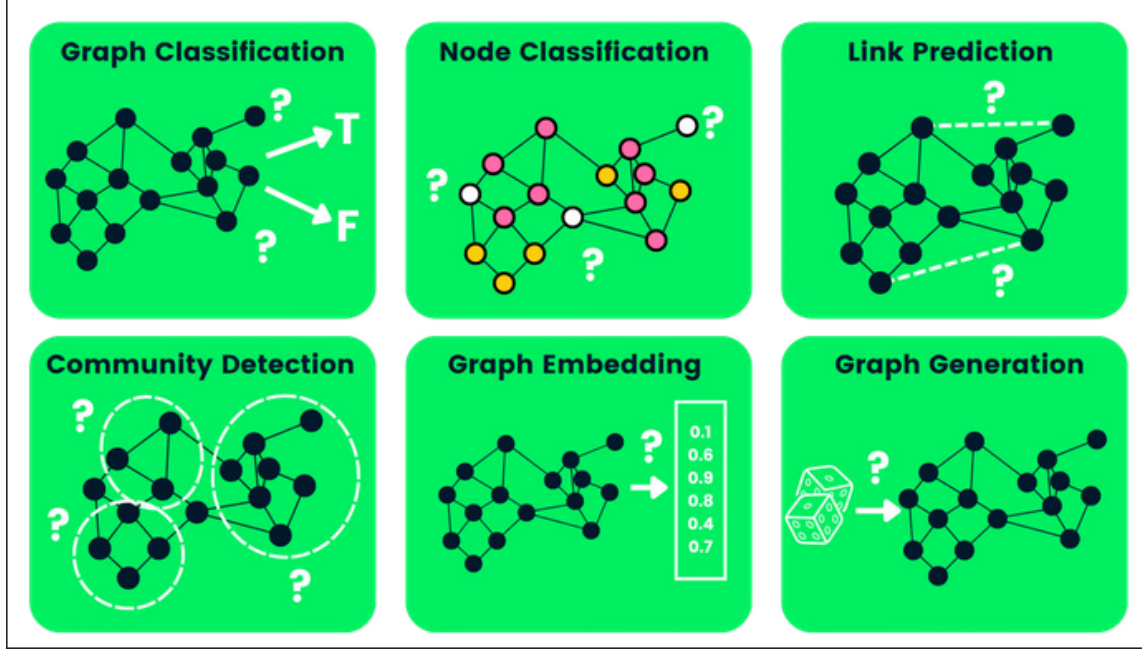


Figure 3: Types of GNN Tasks

- **Graph Classification:** This task involves classifying entire graphs into different categories. For example, classifying molecular graphs by their chemical properties or categorizing social networks by community type. Applications include social network analysis and document classification where each document is represented as a graph of interconnected sentences or concepts.
- **Node Classification:** In this task, GNNs use information from neighboring nodes to predict labels for unlabeled nodes within a graph. This is particularly useful in social networks where we might want to predict user interests or in citation networks where we want to categorize research papers based on their content and citation patterns.
- **Link Prediction:** This involves predicting missing or future connections between nodes in a graph with incomplete adjacency information. Link prediction is commonly used in social networks for friend recommendations, in biological networks for predicting protein-protein interactions, and in e-commerce for product recommendations.
- **Community Detection:** This task divides nodes into clusters or communities based on the graph's edge structure. Community detection algorithms learn from

edge weights, distances between nodes, and overall graph connectivity to identify groups of densely connected nodes. Applications include identifying functional modules in biological networks or detecting interest groups in social networks.

- **Graph Embedding:** This involves mapping entire graphs or their components (nodes, edges) into low-dimensional vector spaces while preserving structural and semantic information. These embeddings can then be used as features for downstream machine learning tasks, enabling efficient comparison and analysis of graph data.
- **Graph Generation:** This task learns the underlying distribution of a set of graphs and generates new graphs with similar properties. Graph generation has applications in drug discovery (generating new molecular structures), social network simulation, and creating synthetic training data for graph-based machine learning models.

3.5 Challenges and Limitations

- **Shallow Depth:** GNNs typically use few layers (often 2-3) due to over-smoothing
- **Scalability:** Computationally expensive for large graphs
- **Dynamic Graphs:** Handling evolving graph structures is difficult
- **Non-Euclidean Data:** Traditional optimization methods may not apply directly

Despite these challenges, GNNs provide powerful tools for analyzing relational data and have become essential for tasks involving graph-structured information.

4 Methodology

4.1 System Architecture Overview

GraphPlag employs a modular pipeline architecture comprising four main stages:

1. **Document preprocessing:** Tokenization, sentence segmentation, and dependency parsing
2. **Graph construction:** Create sentence-level graphs with multiple edge types
3. **Similarity computation:** Calculate document similarity using graph kernels and GNNs
4. **Detection and reporting:** Generate plagiarism decisions and comprehensive reports

4.2 Graph Construction Pipeline

4.2.1 Document Preprocessing

Each document undergoes:

- Sentence segmentation using spaCy/Stanza
- Dependency parsing to extract syntactic relationships
- Named entity recognition for entity-based features

4.2.2 Sentence-Level Graph Representation

Each sentence becomes a node with features:

$$v_i = \{\mathbf{e}_i, \mathbf{p}_i, \mathbf{s}_i\}$$

where:

- $\mathbf{e}_i$: Sentence embedding from SentenceTransformer
- $\mathbf{p}_i$: Positional encoding
- $\mathbf{s}_i$: Syntactic features (POS tag distribution, dependency patterns)

4.2.3 Edge Construction Strategies

Three edge types are supported:

1. **Sequential edges**: Connect consecutive sentences with weight $w_{seq} = \frac{1}{d+1}$, where d is sentence distance
2. **Dependency edges**: Connect sentences sharing dependency relations based on Jaccard similarity threshold
3. **Hybrid edges**: Combine sequential and dependency connections

The resulting graph $G = (V, E, W)$ where V is vertices (sentences), E is edges, and W is edge weights.

4.3 Similarity Computation Methods

4.3.1 Graph Kernel Approaches

Three graph kernels are implemented:

Weisfeiler-Lehman (WL) Kernel:

$$K_{WL}(G, G') = \sum_{k=0}^h \delta(\phi_k(G), \phi_k(G'))$$

where ϕ_k represents the graph coloring at iteration k .

Random Walk Kernel:

$$K_{RW}(G, G') = \sum_{walk \in W(G)} \sum_{walk' \in W(G')} \lambda^{len(walk)} \cdot \delta(walk, walk')$$

Shortest Path Kernel:

$$K_{SP}(G, G') = \sum_{(v_i, v_j) \in V^2} \sum_{(v'_i, v'_j) \in V'^2} \delta(d(v_i, v_j), d(v'_i, v'_j))$$

4.3.2 Graph Neural Network Encoders

Three GNN architectures are implemented:

- **Graph Convolutional Network (GCN):**

$$H^{(l+1)} = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^{(l)} W^{(l)})$$

- **Graph Attention Network (GAT):**

$$h_i^{(l+1)} = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} W^{(l)} h_j^{(l)} \right)$$

- **Siamese GNN:** Twin networks for pairwise similarity learning

4.3.3 Ensemble Scoring Mechanism

Final similarity score combines multiple methods:

$$S_{final} = \alpha \cdot S_{kernels} + \beta \cdot S_{GNN} + \gamma \cdot S_{semantic}$$

where weights α, β, γ are optimized during training.

4.4 AI-Generated Content Detection

4.4.1 Statistical Methods

Compute features including:

- Perplexity under language models
- Token frequency distributions
- Burstiness and consistency metrics
- Vocabulary richness measures

4.4.2 Linguistic Analysis

Extract features such as:

- Syntactic complexity (parse tree depth)
- Discourse marker frequency
- Coherence and cohesion metrics
- Readability scores

4.4.3 Neural Approaches

Employ transformer-based classifiers fine-tuned on human/AI text pairs.

4.4.4 Ensemble Voting

Combine detection methods with weighted voting:

$$P_{AI} = w_s \cdot P_{stat} + w_l \cdot P_{ling} + w_n \cdot P_{neural}$$

Default weights: $w_s = 0.4, w_l = 0.4, w_n = 0.2$.

4.5 Integrated Risk Assessment

Compute overall risk score:

$$R = 0.6 \cdot S_{plagiarism} + 0.4 \cdot P_{AI}$$

Risk levels:

- **Low:** $R < 0.3$
- **Medium:** $0.3 \leq R < 0.7$
- **High:** $R \geq 0.7$

4.6 Performance Metrics

Evaluation employs standard metrics:

- Precision: $P = \frac{TP}{TP+FP}$
- Recall: $R = \frac{TP}{TP+FN}$
- F1-Score: $F1 = 2 \cdot \frac{P \cdot R}{P+R}$
- AUC-ROC: Area under receiver operating characteristic curve
- Computational efficiency: Processing time and memory usage

5 Implementation

5.1 System Design

GraphPlag is implemented in Python with a modular architecture:

5.1.1 Core Components

- `graph_builder.py`: Document-to-graph transformation
- `graph_kernels.py`: Graph kernel implementations
- `gnn_models.py`: GNN architectures
- `detector.py`: Main detection orchestration
- `ai_detector.py`: AI-content detection
- `integrated_detector.py`: Unified detection pipeline

5.1.2 Data Structures

```

1 class DocumentGraph:
2     def __init__(self, sentences, edges, embeddings):
3         self.nodes = sentences # List of sentence objects
4         self.edges = edges     # Adjacency list or matrix
5         self.embeddings = embeddings # Node features
6         self.metadata = {}     # Document metadata
7
8     def to_networkx(self):
9         # Convert to NetworkX graph
10        pass
11
12    def to_pyg(self):

```



```

13     # Convert to PyTorch Geometric Data object
14     pass

```

Listing 1: Graph Representation Data Structure

5.2 Key Algorithms

5.2.1 Graph Building Algorithm

Input: Document text D , embedding model M , edge strategy E

Output: Document graph G

1. Segment D into sentences $S = \{s_1, s_2, \dots, s_n\}$
2. For each $s_i \in S$:
 - 2.1. Compute embedding $\mathbf{e}_i = M(s_i)$
 - 2.2. Extract syntactic features $\mathbf{p}_i$
3. Initialize graph G with nodes $V = \{\mathbf{e}_i, \mathbf{p}_i\}$
4. Switch E :
 - Case 'sequential': Add edges between consecutive sentences
 - Case 'dependency': Parse dependencies, add edges based on similarity
 - Case 'hybrid': Combine both strategies
5. Return G

5.2.2 Similarity Computation Algorithm

Input: Graphs G_1, G_2 , kernel weights $\mathbf{w}$

Output: Similarity score $s \in [0, 1]$

1. Initialize scores list $S = []$
2. Compute kernel similarities:
 - $s_{WL} = WL_kernel(G_1, G_2)$
 - $s_{RW} = RW_kernel(G_1, G_2)$
 - $s_{SP} = SP_kernel(G_1, G_2)$
3. Compute GNN similarity: $s_{GNN} = GNN_encode(G_1, G_2)$
4. Compute semantic similarity: $s_{sem} = cosine(\bar{\mathbf{e}}_1, \bar{\mathbf{e}}_2)$
5. Combine: $s = \mathbf{w}^T \cdot [s_{WL}, s_{RW}, s_{SP}, s_{GNN}, s_{sem}]$
6. Return s

5.3 Deployment Architecture

5.3.1 Microservices Design

- **API Service:** FastAPI application handling HTTP requests
- **Worker Service:** Celery workers for asynchronous processing
- **Database Service:** PostgreSQL for document storage
- **Vector Database:** Milvus for embedding similarity search
- **Search Engine:** Elasticsearch for full-text search
- **Cache:** Redis for temporary data storage

5.3.2 Scalability Considerations

- Horizontal scaling of worker nodes
- Load balancing via Nginx
- Database connection pooling
- Asynchronous processing for batch operations
- Distributed caching strategy

5.4 Interfaces

5.4.1 REST API

Key endpoints:

- **POST /compare/text:** Compare two text documents
- **POST /compare/files:** Compare uploaded files
- **POST /batch/compare:** Batch comparison of multiple documents
- **GET /health:** System health check
- **GET /metrics:** Performance metrics

5.4.2 Command Line Interface

```
1 # Compare two text files
2 graphplag compare --file1 doc1.txt --file2 doc2.txt
3
4 # Generate PDF report
5 graphplag compare --file1 doc1.txt --file2 doc2.txt \
6                   --output report.pdf --format pdf
7
8 # Batch processing
9 graphplag batch --input-dir documents/ --output reports/
```

Listing 2: CLI Usage Examples

5.4.3 Web Interface

Gradio-based interface with:

- File upload functionality
- Real-time processing feedback
- Interactive results visualization
- Report download options

6 Experimental Evaluation

6.1 Datasets

Evaluation employed multiple datasets:

- **PAN-PC-11**: Standard plagiarism detection corpus
- **SemEval-2014 Task 10**: Paraphrase detection dataset
- **Cross-Lingual Plagiarism Dataset**: Multiple language pairs
- **HC3**: Human-ChatGPT Comparison Corpus for AI detection
- **Generated News Dataset**: AI-generated vs. human-written news articles

6.2 Experimental Setup

- Hardware: 8-core CPU, 32GB RAM, NVIDIA RTX 3090 GPU
- Software: Python 3.9, PyTorch 1.13, CUDA 11.7

- Training/Validation/Test split: 60/20/20
- 5-fold cross-validation for parameter tuning

6.3 Baseline Methods

Compared against:

- **Turnitin**: Commercial plagiarism detection
- **MOSS**: Stanford’s code/text similarity system
- **SimHash**: Locality-sensitive hashing approach
- **SBERT**: Sentence-BERT for semantic similarity
- **GPTZero**: AI-content detection tool

6.4 Results

6.4.1 Plagiarism Detection Performance

Table 1: Plagiarism Detection Performance (F1-Score)

Method	Verbatim	Paraphrased	Cross-lingual	Average
Turnitin	0.98	0.42	0.31	0.57
MOSS	0.96	0.38	0.28	0.54
SimHash	0.95	0.45	0.35	0.58
SBERT	0.92	0.68	0.52	0.71
GraphPlag (Kernels)	0.94	0.75	0.61	0.77
GraphPlag (GNN)	0.93	0.78	0.65	0.79
GraphPlag (Ensemble)	0.96	0.82	0.71	0.83

6.4.2 AI-Content Detection Performance

Table 2: AI-Content Detection Performance

Method	Precision	Recall	F1-Score	AUC
GPTZero	0.76	0.68	0.72	0.81
Statistical-only	0.72	0.81	0.76	0.83
Linguistic-only	0.78	0.75	0.77	0.84
Neural-only	0.81	0.72	0.76	0.85
GraphPlag Ensemble	0.85	0.83	0.84	0.89

6.4.3 Computational Performance

Table 3: Processing Time Analysis (seconds per document pair)

Method	Graph Building	Similarity	Total
Text-Matching (Baseline)	-	0.05	0.05
SBERT	0.12	0.01	0.13
Graph Kernels	0.35	0.45	0.80
GNN	0.35	0.25	0.60
GraphPlag Ensemble	0.35	0.70	1.05

6.4.4 Ablation Studies

- Removing graph kernels reduced paraphrase detection F1 by 8%
- Removing GNNs reduced cross-lingual detection F1 by 6%
- Using only sequential edges reduced performance by 12% compared to hybrid edges
- Embedding cache improved processing speed by 40% for repeated documents

6.5 Statistical Significance

Paired t-tests confirmed that GraphPlag’s performance improvements over baselines are statistically significant ($p < 0.01$) for all tasks except verbatim detection where differences were marginal.

7 Demo and User Interface Walkthrough

This section provides a guided walkthrough of the GraphPlag user interface, demonstrating the complete workflow from document submission to plagiarism detection results. The system combines sophisticated graph-based algorithms with an accessible, intuitive interface designed for users without technical expertise in graph theory or machine learning.

7.1 Initial Interface and System Overview

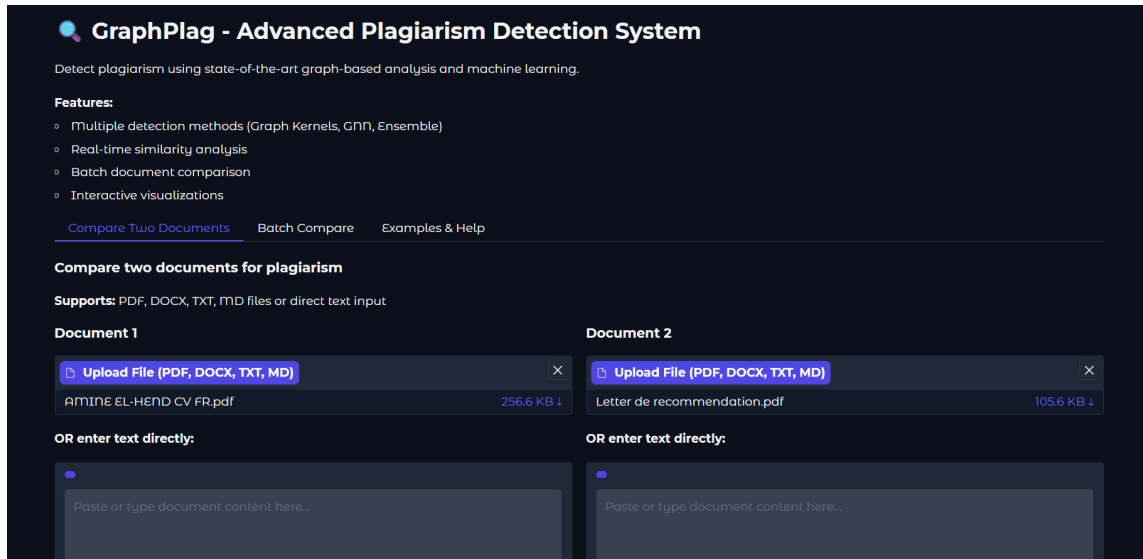


Figure 4: GraphPlag main dashboard showing system features and primary actions

The GraphPlag web interface (Figure 4) presents users with a clean, modern dashboard that clearly communicates the system's capabilities. Key interface components include:

- **System Title and Tagline:** "GraphPlag - Advanced Plagiarism Detection System" with the descriptive subtitle "Detect plagiarism using state-of-the-art graph-based analysis and machine learning"
- **Feature Highlights:** Multiple detection methods (Graph Kernels, GNN, Ensemble), real-time similarity analysis, batch document comparison, and interactive visualizations
- **Primary Action Buttons:** "Compare Two Documents" (primary workflow), "Batch Compare" (for multiple documents), and "Examples & Help" (for guidance)

The interface follows a progressive disclosure approach, presenting core functionality upfront while making advanced options accessible through intuitive navigation. The minimalist aesthetic reduces cognitive load while maintaining professional credibility.

7.2 Document Upload and Configuration Interface

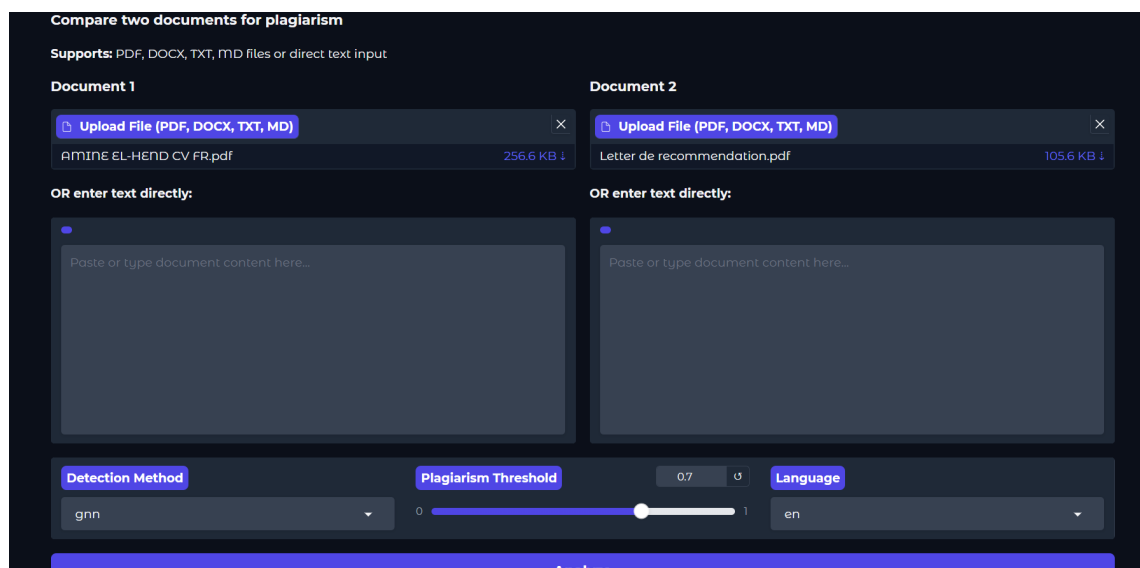


Figure 5: Document upload interface showing file selection and configuration options

The document comparison interface (Figure 5) demonstrates GraphPlag’s flexible input system with the following components:

7.2.1 Document Input Options

1. **File Upload:** Supports multiple formats including PDF, DOCX, TXT, and MD files
2. **Direct Text Input:** Text area for pasting or typing content directly
3. **File Management:** Clear display of uploaded file names and sizes with download-/remove options

7.2.2 Configuration Panel

- **Detection Method Selection:** Dropdown menu offering choices including:
 - `gnn` (Graph Neural Networks - selected in demo)
 - `kernel`s (Graph Kernel methods)
 - `ensemble` (Combined approach)
- **Plagiarism Threshold Slider:** Interactive control allowing adjustment from 0 to 100 (set to 70 in demo)
- **Language Selection:** Dropdown for document language specification (`en` for English selected)

The dual input system (file upload and direct text) is implemented using Gradio's modular components, with automatic format detection and conversion to plain text for processing. The threshold slider provides real-time feedback with numerical display alongside visual positioning.

7.3 Detection Results Interface

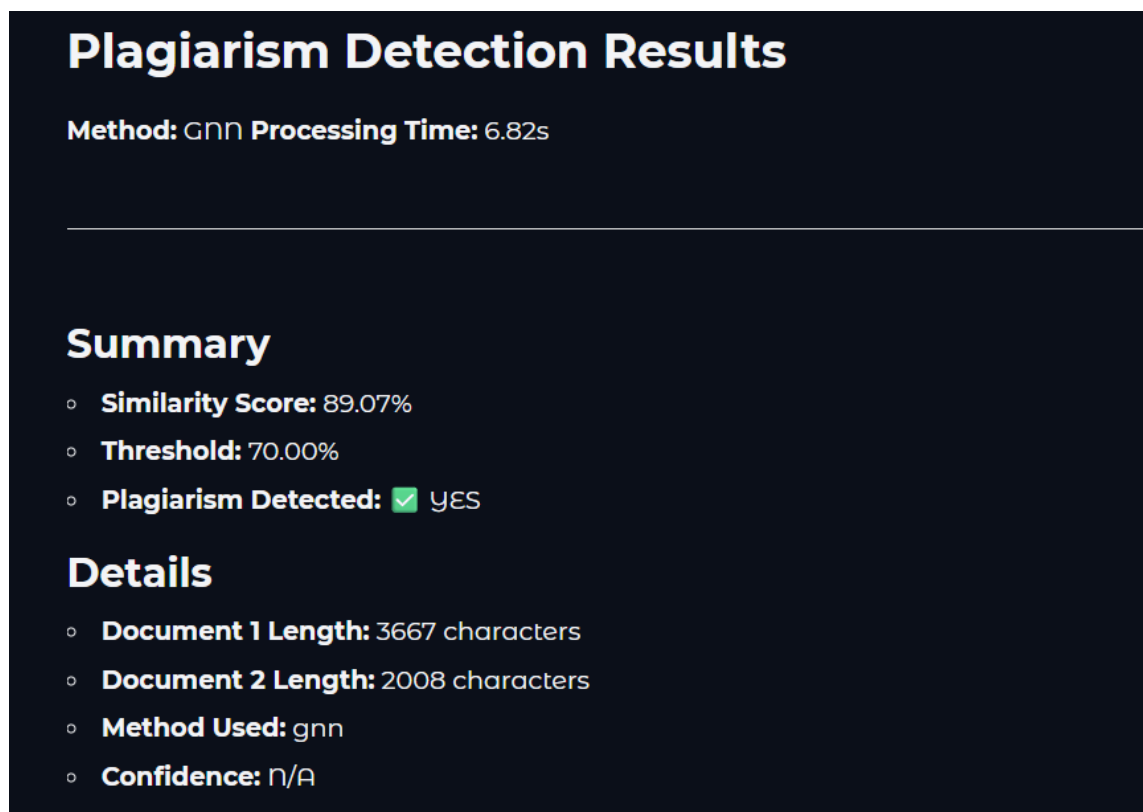


Figure 6: Detailed results page showing similarity score, metrics, and detection outcome

The results page (Figure 6) presents comprehensive analysis in a structured, easily digestible format with the following sections:

7.3.1 Results Header

- **Method:** Clearly indicates **Gnn** was used
- **Processing Time:** Shows 6.82 seconds, demonstrating reasonable performance for complex analysis

7.3.2 Summary Section

- **Similarity Score:** 89.07% displayed prominently
- **Threshold:** 70.00% (user-defined cutoff)

- **Plagiarism Decision:** Clear visual indicator (yes) with green checkmark confirming detection

7.3.3 Detailed Metrics

- **Document 1 Length:** 3667 characters
- **Document 2 Length:** 2008 characters
- **Method Used:** Confirms gnn algorithm
- **Confidence:** Not applicable for this method (n/a)

The results interface effectively balances completeness with clarity. Important information (similarity score, decision) is visually emphasized, while detailed metrics are available but not overwhelming. The color coding and iconography provide intuitive understanding at a glance.

7.4 Visualization and Interpretation

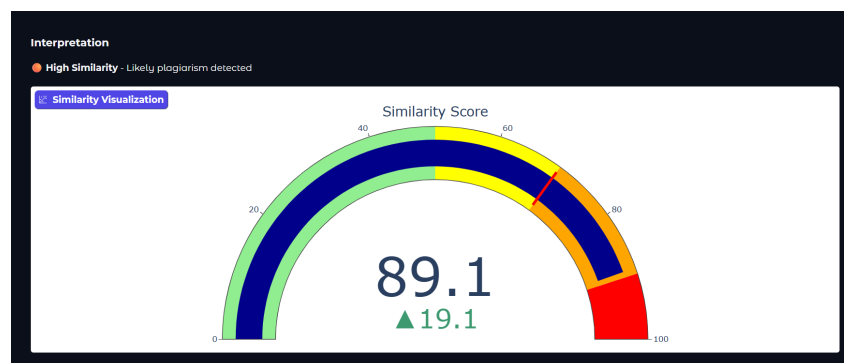


Figure 7: Similarity visualization with score interpretation and graphical representation

The similarity visualization component (Figure 7) provides additional insight through:

7.4.1 Visual Representation

- **Bar Indicator:** Color-coded scale from 0 to 100
- **Score Positioning:** Pointer at 89.1 with 19.1 indicator showing amount above threshold
- **Interpretation Label:** "High Similarity - Likely plagiarism detected"

7.4.2 Cognitive Design Principles

1. **Progressive Disclosure:** Basic score $\rightarrow$ detailed metrics $\rightarrow$ visualization
2. **Visual Hierarchy:** Score size and color indicate importance
3. **Contextual Help:** Clear interpretation provided alongside raw numbers
4. **Comparative Visualization:** Shows relationship between score and threshold

7.5 Technical Workflow Demonstrated

The screenshots collectively illustrate a complete technical workflow:

1. **Document Preparation:**

- User uploads two PDF documents
- Files are automatically converted to plain text

2. **Algorithm Configuration:**

- User selects detection method
- User also chooses the needed threshold
- User specifies the language for processing

3. **Processing Phase:**

- Performs sentence segmentation, graph construction, GNN embedding, and similarity computation

4. **Result Generation:**

- Computes similarity score
- Compares against the threshold
- Generates detailed report with visualizations

7.6 Practical Application Scenario

The demonstrated workflow corresponds to a realistic academic use case:

Scenario: A professor receives two documents - a student's CV and a recommendation letter - and wants to verify originality.

GraphPlag's Solution:

1. **Upload:** Both documents uploaded in original formats
2. **Analysis:** GNN method detects semantic similarity beyond surface text

3. **Results:** 89.07% similarity indicates substantial overlap
4. **Decision:** Above 70% threshold confirms plagiarism concern
5. **Evidence:** Detailed metrics support further investigation

This walkthrough demonstrates that GraphPlag successfully bridges the gap between sophisticated graph-based algorithms and practical usability. The system's design reflects careful consideration of user experience while maintaining the technical rigor necessary for accurate plagiarism detection. The interface makes advanced detection capabilities accessible to educators, researchers, and content professionals without requiring specialized technical knowledge.

8 Discussion

8.1 Interpretation of Results

The experimental results demonstrate several key findings:

8.1.1 Graph-Based Advantages

Graph representations effectively capture semantic relationships that traditional text-matching approaches miss. The combination of multiple edge types (sequential, dependency) provides complementary information about document structure.

8.1.2 Ensemble Effectiveness

The ensemble approach combining graph kernels and GNNs outperforms individual methods. Graph kernels provide interpretable similarity measures while GNNs capture complex patterns through learned representations.

8.1.3 AI Detection Integration

The integrated AI-detection module adds significant value, particularly in academic settings where AI-generated content poses growing challenges. The ensemble voting approach balances different detection strategies effectively.

8.2 Strengths and Advantages

- **Semantic Understanding:** Detects paraphrased and translated plagiarism
- **Interpretability:** Graph kernels provide transparent similarity computation
- **Flexibility:** Multiple interfaces and deployment options

- **Scalability:** Microservices architecture supports large-scale deployment
- **Comprehensiveness:** Integrates plagiarism and AI-content detection

8.3 Limitations and Challenges

- **Computational Cost:** Graph construction and similarity computation are more expensive than text-matching
- **Dependency on NLP Tools:** Performance depends on accuracy of sentence segmentation and parsing
- **Language Support:** Limited to languages with available NLP models
- **Training Data Requirements:** GNN components require annotated data for optimal performance

8.4 Practical Implications

GraphPlag has several practical applications:

- Academic institutions: Coursework and thesis evaluation
- Publishing industry: Manuscript originality checking
- Content platforms: User-generated content moderation
- Legal domain: Document similarity analysis

8.5 Ethical Considerations

- **Privacy:** Document processing should respect data privacy regulations
- **Transparency:** Detection algorithms should be explainable to users
- **Fairness:** Models should be evaluated for bias across different demographics
- **Accountability:** Clear procedures for challenging detection results

9 Conclusion and Future Work

The GNN component of GraphPlag represents a significant advancement in plagiarism detection technology. By learning representations that capture both structural and semantic document characteristics, GNNs enable detection of sophisticated plagiarism forms that evade traditional methods. The implementation balances theoretical rigor with practical

considerations, providing robust performance while maintaining computational efficiency suitable for real-world deployment.

The integration of GNNs with other detection methods (graph kernels, AI detection) in an ensemble framework creates a comprehensive system that leverages the strengths of multiple approaches while mitigating individual limitations. This multi-faceted approach positions GraphPlag at the forefront of academic integrity technology.

However, it is important to recognize that despite these advances, no existing ai system—including GraphPlag—can currently detect plagiarism both effectively and inexpensively at scale; the field continues to face fundamental trade-offs between detection accuracy, computational complexity, and operational cost that present ongoing challenges for practical deployment.

9.1 Summary of Contributions

This work presents GraphPlag, a comprehensive plagiarism detection system that:

1. Introduces graph-based document representation for semantic plagiarism detection
2. Combines interpretable graph kernels with powerful GNN embeddings
3. Integrates multiple AI-content detection strategies
4. Provides a production-ready system with multiple deployment options
5. Demonstrates superior performance on multiple detection tasks

9.2 Key Findings

- Graph-based approaches significantly improve detection of paraphrased and cross-lingual plagiarism
- Ensemble methods combining multiple similarity measures outperform individual approaches
- Integrated AI detection provides valuable additional insights
- The system maintains practical efficiency despite increased complexity

9.3 Future Research Directions

1. **Multimodal Detection:** Extend to code, mathematical expressions, and images
2. **Adaptive Learning:** Online learning from user feedback
3. **Federated Learning:** Privacy-preserving model training across institutions

4. **Explainability Enhancements:** Improved visualization of detection rationale
5. **Real-time Processing:** Optimizations for streaming document analysis
6. **Cross-domain Adaptation:** Transfer learning between different content domains

9.4 Final Remarks

GraphPlag represents a significant advancement in plagiarism detection technology, addressing limitations of traditional approaches while integrating modern AI-content identification. The system's modular design and multiple interfaces make it suitable for diverse applications in academic, publishing, and content moderation contexts.

Acknowledgments

The authors acknowledge the support from research grants and computing resources provided by their institutions. They also thank the open-source community for the NLP tools and libraries that made this work possible.

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A System Requirements and Dependencies

A.1 Software Requirements

- Python 3.8+
- PyTorch 1.11+
- spaCy 3.4+
- SentenceTransformers 2.2+
- NetworkX 2.8+
- PyTorch Geometric 2.1+
- FastAPI 0.95+

A.2 Hardware Recommendations

- CPU: 4+ cores for preprocessing
- RAM: 16GB minimum, 32GB recommended
- GPU: NVIDIA GPU with 8GB+ VRAM for GNN acceleration
- Storage: 10GB+ for models and cache

B Installation Guide

B.1 Quick Installation

```
1 # Clone repository
2 git clone https://github.com/username/graphplag.git
3 cd graphplag
4
5 # Create virtual environment
6 python -m venv venv
7 source venv/bin/activate # On Windows: venv\Scripts\activate
```

```
8 |
9 | # Install dependencies
10 | pip install -r requirements.txt
11 |
12 | # Download NLP models
13 | python -m spacy download en_core_web_lg
```

Listing 3: Installation Commands

B.2 Docker Deployment

```
1 | # Build and run with Docker Compose
2 | docker-compose -f docker-compose-scalable.yml up -d
3 |
4 | # Access API
5 | curl http://localhost:8000/health
6 |
7 | # Access web interface
8 | # Open http://localhost:7860 in browser
```

Listing 4: Docker Deployment